

## Secant substitution — 3.3

Example: Find  $\int \frac{\sqrt{4x^2-1}}{x} dx$

$$4x^2-1 = 4(x^2 - \frac{1}{4})$$

$$x = \frac{1}{2} \sec \theta$$

$$dx = \frac{1}{2} \sec \theta \tan \theta d\theta$$

$$= \int \frac{\sqrt{4 \cancel{\frac{1}{4}} \sec^2 \theta - 1}}{\cancel{\frac{1}{2}} \sec \theta} \cdot \cancel{\frac{1}{2}} \sec \theta \tan \theta d\theta$$

$$= \int \sqrt{\sec^2 \theta - 1} \cdot \tan \theta d\theta = \int |\tan \theta| \cdot \tan \theta d\theta$$

$$= \begin{cases} \int \tan^2 \theta d\theta & \tan \theta \geq 0 & \text{Q I} & x \geq \frac{1}{2} \\ -\int \tan^2 \theta d\theta & \tan \theta \leq 0 & \text{Q II} & x \leq -\frac{1}{2} \end{cases}$$

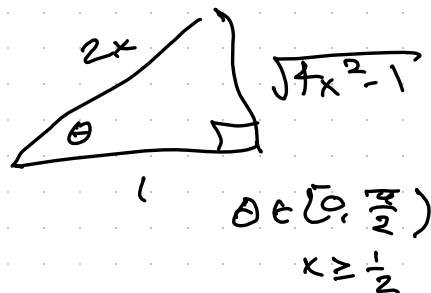
$$\int \tan^2 \theta d\theta = \int \sec^2 \theta - 1 d\theta = \tan \theta - \theta + C$$

$$= \begin{cases} \tan \theta - \theta & x \geq \frac{1}{2} \quad / \quad \theta \text{ is in QI} \\ \theta - \tan \theta & x \leq -\frac{1}{2} \quad / \quad \theta \text{ is in QII} \end{cases}$$

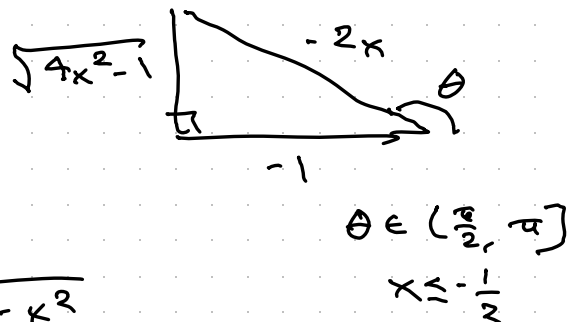
$$2x = \sec \theta$$

$$\theta = \operatorname{arcsec}(2x)$$

$$= \begin{cases} \sqrt{4x^2 - 1} - \operatorname{arcsec}(2x) & x \geq \frac{1}{2} \\ \operatorname{arcsec}(2x) - (-\sqrt{4x^2 - 1}) & x \leq -\frac{1}{2} \end{cases}$$



$$= \sqrt{4x^2 - 1} + \begin{cases} -\operatorname{arcsec}(2x) & x \geq \frac{1}{2} \\ \operatorname{arcsec}(2x) & x \leq -\frac{1}{2} \end{cases}$$



$$\sqrt{x^2 - a^2}$$

$$a \sec(\theta)$$

$$\sqrt{a^2 + x^2}$$

$$a \tan \theta$$

$$\sqrt{a^2 - x^2}$$

$$a \sin \theta$$